Med Sum-AI:

Intelligent Clinical Report Summarization and Predictive Outcome Analytics

Using Transformer-Based NLP & Advanced Statistical Models

Capstone Project Synopsis

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1. Introduction

1.1 Background and Context

Healthcare systems worldwide generate over 3.5 billion clinical notes annually, yet clinicians spend an estimated 35–55% of their working hours on documentation rather than direct patient care. This imbalance represents a systemic crisis that drives clinician burnout, increases the probability of medical error, and erodes the quality of patient-clinician interaction. A single intensive care unit (ICU) discharge summary may span 3,000 words; a radiologist reviews 70–100 reports per shift; an attending physician on a 12-hour call inherits 20 or more patient handover notes from the preceding team. The cumulative cognitive load is not sustainable, and current electronic health record (EHR) systems offer no meaningful relief.

At the same time, the unstructured text contained within these clinical documents harbors untapped prognostic value. Discharge summaries encode nuanced clinical judgments, treatment trajectories, and comorbidity narratives that structured EHR fields — International Classification of Diseases (ICD) codes, vital signs, laboratory values — can only partially capture. Studies demonstrate that incorporating natural language processing (NLP)-derived features from clinical notes improves predictive model performance for 30-day hospital readmission by 5–12% over structured-data-only baselines (Adams et al., 2021). Despite this evidence, the majority of deployed outcome prediction systems continue to ignore the textual record entirely.

The transformer revolution in NLP — inaugurated by the self-attention architecture of Vaswani et al. (2017) and accelerated by domain-specific pre-training in models such as Bio BERT (Lee et al., 2020), Clinical BERT (Alsentzer et al., 2019), and BART (Lewis et al., 2020) — has created, for the first time, the technical infrastructure needed to process clinical text at scale with both linguistic fluency and factual accuracy. However, clinical NLP systems face a persistent and patient-safety-critical challenge: abstractive summarization models generate factually unsupported statements (hallucinations) at rates of 10–18% in clinical settings (Adams et al., 2021). No published work has systematically benchmarked multiple transformer models on the same dataset, measured and mitigated hallucination rates with clinical rigor, and simultaneously used the generated summaries as inputs to downstream outcome prediction models.

The United Arab Emirates National AI Strategy 2031 and the Dubai Health Authority Digital Health Strategy both prioritize AI adoption in healthcare, creating immediate national and regional relevance for this research. The availability of three large, publicly accessible clinical datasets — CASIA-CXR (www.casia-cxr.net), the i2b2/n2c2 NLP Challenge corpus, and the Indiana University Chest X-Ray dataset — provides the empirical foundation to conduct this study rigorously and reproducibly.

1.2 Problem Statement

Clinical reports are verbose, structurally inconsistent, and written in domain-specific medical language that general-purpose NLP models fail to process accurately. Manual summarization is time-consuming, prone to critical omissions, and inconsistent across clinicians and institutions. Existing automated summarization tools either rely on outdated rule-based approaches or produce outputs with unacceptably high rates of clinically dangerous hallucinations. Furthermore, the substantial prognostic richness of unstructured clinical text is almost entirely untapped by current

patient outcome prediction systems, leaving diagnostic errors and pathology misclassification unaddressed. No unified system currently combines transformer-based summarization, hallucination mitigation, NLP-derived outcome prediction, and explainability in a single validated pipeline.

1.3 Purpose of the Study

This study aims to develop and validate Med Sum-AI — a unified transformer-NLP and statistical modelling pipeline that (1) automatically generates clinically accurate abstractive and extractive summaries of radiology reports by fine-tuning and benchmarking BERT, BioBERT, ClinicalBERT, and BART; (2) detects and mitigates hallucinations using Retrieval-Augmented Generation (RAG) and constrained decoding; (3) predicts chest X-ray pathology outcomes — classification across five CASIA-CXR categories (Cardiomegaly, Pneumonia, Pneumothorax, Pleural Effusion, Mass) — from NLP-extracted structured features using Logistic Regression, Random Forest, XGBoost, and Cox Proportional Hazard Models; and (4) makes all system decisions interpretable to clinicians through SHAP token attribution and LIME feature importance dashboards. All components are validated with automated NLP metrics, clinical expert assessment, and statistical significance testing across three publicly available datasets.

2. Scope and Objectives

2.1 Project Scope

Med Sum-AI is bounded by the following scope parameters. Input data consists of de-identified chest X-ray radiology reports and paired images from the CASIA-CXR dataset (www.casia-cxr.net), covering five pathology categories — Cardiomegaly, Pneumonia, Pneumothorax, Pleural Effusion, and Mass — supplemented by the i2b2/n2c2 NLP Challenge corpus for generalizability testing and the Indiana University Chest X-Ray corpus for domain transfer evaluation. NLP models evaluated include BERT (extractive baseline via BERT Sum), BioBERT, ClinicalBERT, and BART (abstractive), with a voting ensemble as the final production layer. Predictive models include Logistic Regression, Random Forest, XGBoost, and the Cox Proportional Hazard Model. Evaluation encompasses automated NLP metrics (ROUGE-1/2/L, BERTScore, Fact CC, QuestEval), a five-dimension clinical expert rubric, and predictive performance metrics (AUC-ROC, C-statistic, Brier Score). Out of scope are real-time EHR deployment, proprietary hospital records, and non-English clinical text. The project is feasible within 10 weeks: Week 1–2 covers data acquisition and preprocessing; Week 3–4 covers NLP model fine-tuning (RQ1); Week 5 covers clinical expert evaluation (RQ2); Week 6 covers hallucination experiments (RQ3); Week 7–8 covers predictive modelling (RQ4); Week 9 covers explainability and trust evaluation (RQ5); Week 10 covers cross-domain testing (RQ6) and final reporting.

2.2 Objectives

- Obj. 1: Build a multi-model NLP pipeline for abstractive and extractive summarization of CASIA-CXR radiology reports across the five pathology categories (Cardiomegaly, Pneumonia, Pneumothorax, Pleural Effusion, Mass).

- Obj. 2: Fine-tune ClinicalBERT and BART on CASIA-CXR radiology reports and evaluate against ROUGE, BERTScore, and Fact CC baselines.
- Obj. 3: Engineer 20 structured feature variables from CASIA-CXR radiology report text and label files for downstream statistical outcome prediction.
- Obj. 4: Develop and compare Logistic Regression, Random Forest, XGBoost, and Cox PHM for chest X-ray pathology classification across the five CASIA-CXR disease categories.
- Obj. 5: Validate all models with appropriate statistical tests (ANOVA, Kruskal-Wallis, McNemar's, DeLong) and structured expert clinical review.
- Obj. 6: Deliver SHAP/LIME explainability dashboards and a fully open-source GitHub repository with Docker containerized environment.

2.3 Research Questions

The central research problem is: How can a unified transformer-NLP and statistical modelling pipeline automatically summarize radiology reports and predict chest X-ray pathology outcomes with clinically meaningful accuracy — and how can the system's decisions be made explainable to clinicians? This problem is operationalized through four primary research questions that form the basis of the sample size calculations, analytic approach, and evaluation framework.

Research Question 1 (RQ1): NLP Model Benchmarking

What is the comparative summarization performance of BERT, BioBERT, ClinicalBERT, and BART when fine-tuned on CASIA-CXR radiology reports, as measured by ROUGE-1/2/L and BERTScore-F1?

Research Question 2 (RQ2): Clinical Expert Validation

To what extent do abstractive summaries generated by BART and ClinicalBERT achieve clinically acceptable quality across five dimensions — factual accuracy, completeness, conciseness, clinical relevance, and readability — as assessed by board-certified clinicians using a validated 5-point Likert rubric?

Research Question 3 (RQ3): Hallucination Detection and Mitigation

What proportion of abstractive summaries contain clinically dangerous hallucinations, and which mitigation strategy — constrained decoding, copy mechanism, or Retrieval-Augmented Generation (RAG) most effectively reduces hallucination rate as measured by Fact CC, QuestEval, and manual clinical annotation?

Research Question 4 (RQ4): Predictive Outcome Modelling

Can structured features extracted from NLP-generated radiology report summaries predict chest X-ray pathology outcomes — classification across five CASIA-CXR disease categories (Cardiomegaly, Pneumonia, Pneumothorax, Pleural Effusion, Mass) — with clinically acceptable performance (AUC-ROC > 0.80, F1 > 0.78) using Logistic Regression, Random Forest, XGBoost, and Cox Proportional Hazard Models?

Research Question 5 (RQ5): Explainability and Clinician Trust

Does exposing clinicians to SHAP token attribution maps and LIME feature importance explanations significantly increase their trust in Med Sum-AI’s summarization and prediction outputs, as measured by a validated Technology Acceptance Model (TAM) survey instrument?

Research Question 6 (RQ6): Cross-Domain Generalizability

To what extent do the Med Sum-AI transformer models fine-tuned on CASIA-CXR radiology reports generalize to unseen clinical text domains — specifically the i2b2/n2c2 NLP Challenge corpus and the Indiana University Chest X-Ray dataset — as measured by ROUGE-L Generalisation Index (GI)?

3. Sample Size Calculation

Sample sizes for each research question were determined using either power analysis (for hypothesis testing) or confidence interval width criteria (for estimation studies). The calculations follow the principle that the final adopted sample size must be the maximum of the minimum sample sizes across all research questions to ensure adequate statistical power for the entire study.

Table 1. Sample Size Calculations by Research Question

Research Question	Method Used	Key Parameters	Min. Sample Size (N)
RQ1 — NLP Model Benchmarking	Power Analysis (ANOVA)	$\alpha = 0.05$, Power = 0.80, k = 4 groups, Cohen's f = 0.30	n = 45/group; Project uses N = 2,000 (500/model)
RQ2 — Clinical Expert Validation	Confidence Interval	Z = 1.96, $\sigma = 0.80$, E = ±0.15, estimated mean = 3.5/5.0	n ≥ 109; Project uses N = 240 (60/model × 3 raters)
RQ3 — Hallucination Detection	Power Analysis (McNemar's test)	$\alpha = 0.05$, Power = 0.80, discordant pair proportion = 0.20	n ≥ 150 annotated summaries; Project uses N = 150
RQ4 — Outcome Prediction (LR)	Power Analysis (AUC/Logistic)	AUC = 0.80, null AUC = 0.50, $\alpha = 0.05$, Power = 0.80; EPV = 10, p = 25	n ≥ 250 cases/class per pathology category; Cox PHM ≥ 250 events; CASIA-CXR provides 3,996+ labelled cases

RQ1 requires comparison of four NLP model groups using one-way ANOVA. Using G*Power with alpha = 0.05, power = 0.80, k = 4 groups, and Cohen's f = 0.30 (medium-large effect, consistent with published clinical NLP benchmarks), the minimum sample per group is n = 45. The project uses n = 500 test samples per model (N = 2,000 total) to support robust subgroup analyses and bootstrapped confidence intervals.

RQ2 requires estimation of mean clinical rubric scores with a precision of ±0.15 points at 95% confidence. Using the formula $n = (Z \times \sigma / E)^2$ with Z = 1.96, estimated $\sigma = 0.80$, and E = 0.15, the minimum is n ≥ 109 summaries. The project targets 240 expert-rated summaries (60 per model, three independent raters) to exceed this minimum and enable subgroup analyses by specialty.

RQ3 requires detection of a meaningful difference in hallucination proportions using McNemar's test. With alpha = 0.05, power = 0.80, and an expected discordant pair proportion of 0.20, the

minimum annotation sample is n ≥ 150 summaries. The project uses exactly 150 manually annotated summaries, meeting this requirement.

RQ4 involves logistic regression for binary classification (pathology category prediction). Using the Hanley-McNeil formula for AUC significance (expected AUC = 0.80, null AUC = 0.50, alpha = 0.05, power = 0.80), the minimum is n ≥ 250 cases per pathology category. For the Cox Proportional Hazard Model, applying the events-per-variable (EPV) criterion of 10 with 25 predictors requires a minimum of 250 events. CASIA-CXR provides 3,996 labelled radiology cases across five pathology categories, with each category containing sufficient positive-label instances to meet the minimum event requirement of 250 per class.

Final adopted sample size: N = 2,000 test samples for RQ1 NLP evaluation (maximum across all RQs), with CASIA-CXR's 3,996 labelled radiology cases available for RQ4 predictive modelling across five pathology categories. This sample exceeds the minimum requirements for all four research questions.

4. Data Description

4.1 Primary Dataset: CASIA-CXR Open Chest X-Ray Dataset

CASIA-CXR (Chinese Academy of Sciences Institute of Automation Chest X-Ray) is an open chest X-ray dataset created specifically for benchmarking automatic radiology report generation. It is freely accessible at https://www.casia-cxr.net with no credentialing, institutional approval, or Data Use Agreement required. The dataset contains chest X-ray images and expert-written radiology reports across five clinically important pathological categories: Cardiomegaly, Pneumonia, Pneumothorax, Pleural Effusion, and Mass. For each category, CASIA-CXR provides an Images folder of X-ray images, a Reports CSV file containing full radiology report text (Findings and Impressions sections), and a Labels CSV file with binary pathology classification flags. The total dataset contains 3,996 labelled cases and was purpose-designed for NLP benchmarking, making ROUGE, BERTScore, and FactCC evaluation results directly comparable to published international standards.

The key files used in Med Sum-AI are: the Reports CSV per category (NLP model input; contains the full radiology report text with Findings and Impressions fields), and the Labels CSV per category (provides binary pathology classification labels per X-ray image, used directly as outcome prediction targets for RQ4).

4.2 Secondary and Tertiary Datasets

The i2b2/n2c2 NLP Challenge dataset (https://portal.dbmi.hms.harvard.edu/projects/n2c2-nlp/) is used for cross-domain generalizability testing in RQ6. It contains de-identified clinical notes from Partners Healthcare and is accessible under a Data Use Agreement via the i2b2/n2c2 portal. The Indiana University Chest X-Ray Reports dataset (Open-I, NIH; https://openi.nlm.nih.gov/) contains 3,996 radiology reports with structured findings and impressions and is used for radiology-specific domain transfer evaluation in RQ6. All three datasets are publicly accessible, ethically approved, de-identified, and require no primary data collection by the researcher.

4.3 CASIA-CXR: Dataset Benefits and Justification

MIMIC-III is not available for use in this project. Med Sum-AI therefore adopts the CASIA-CXR (Chinese Academy of Sciences Institute of Automation Chest X-Ray) dataset as its primary empirical foundation. CASIA-CXR is a publicly available, benchmark-designed chest X-ray dataset covering five major pathological categories: Cardiomegaly, Pneumonia, Pneumothorax, Pleural Effusion, and Mass. It can be accessed at www.casia-cxr.net and provides the following key benefits for Med Sum-AI:

- Freely accessible with no credentialing barriers: Unlike MIMIC-III, which requires CITI training and PhysioNet institutional credentialing, CASIA-CXR is directly downloadable from www.casia-cxr.net at no cost and with no special approvals. This removes the single largest access obstacle for this project.
- Clinically rich and multi-category pathology coverage: CASIA-CXR provides labelled data for five distinct thoracic pathologies. Each category includes chest X-ray images, radiology reports (with Findings and Impressions), and structured classification label CSV files, giving Med Sum-AI a diverse, multi-specialty training corpus without requiring institutional data access.
- Direct benchmark alignment for NLP evaluation: CASIA-CXR was designed specifically for automatic radiology report generation benchmarking. ROUGE-1/2/L, BERTScore-F1, FactCC, and QuestEval metrics computed on CASIA-CXR reports align with the evaluation targets established in Section 5 of the Synopsis (Table 3) and the six research questions, without requiring metric recalibration.
- Pre-labelled prediction targets simplify RQ4 engineering: Each CASIA-CXR pathology category ships with a Labels CSV file containing binary pathology classification flags. These serve directly as outcome prediction labels for RQ4, replacing the complex ICD-9 code derivation and ADMISSIONS table joins required in MIMIC-III.
- Supports coherent cross-domain generalizability (RQ6): As a radiology-native dataset, CASIA-CXR enables a more clinically coherent domain transfer evaluation to Indiana University CXR and i2b2 radiology subsets than MIMIC-III's ICU discharge setting would have provided, improving the validity of the RQ6 Generalisation Index.
- Fully de-identified and ethically approved: CASIA-CXR data are de-identified in compliance with international research standards, satisfying the same ethical requirements as MIMIC-III but without any Data Use Agreement or credentialing overhead.

4.4 Data Dictionary — CASIA-CXR Key Variables

Table 2. CASIA-CXR Data Dictionary for Med Sum-AI

Table. Column	Type	Range/Values	Description	Role in Project
REPORTS.TEXT	TEXT	Free text	Full clinical note content	Primary NLP model input
REPORTS.CATEGORY	VARCHAR	Radiology (all cases)	Report section identifier	Filter by report type

LABELS.PATHOLOGY_FLAG	BINARY	0/1	Binary pathology presence flag (1 = condition present)	Pathology classification label (RQ4 outcome)
REPORTS.FINDING_COUNT	FLOAT	0–30 (integer count)	Number of clinical findings mentioned in the Findings section	NLP-derived numeric feature for RQ4 regression
LABELS.CONDITION_LABEL	BINARY	0/1 (derived per category)	Multi-label pathology co-occurrence flag (engineered)	Secondary pathology classification label
REPORTS.FINDINGS_TEXT	VARCHAR	Free text	Full text of the Findings section of the radiology report	Secondary NLP input for feature extraction
LABELS.CATEGORY_NAME	VARCHAR	Categorical string	Pathology category name (Cardiomegaly, Pneumonia, etc.)	Category filter and stratification variable
IMAGES.FILE_PATH	VARCHAR	String (relative path)	File path to the paired chest X-ray image for each report	Image input for multimodal analysis (optional RQ4 extension)

4.5 GitHub Repository and Reproducibility

All preprocessing code, model training scripts, evaluation notebooks, SHAP/LIME dashboards, and experimental results will be published at: https://github.com/AsmaaHamedEwes/MedSum-AI. The repository will include a fully reproducible Docker containerized environment specifying all library versions, random seeds, and data split indices to enable complete reproduction of all reported results. The repository will be publicly accessible under the MIT license prior to the final submission deadline.

5. Analytic Approach

5.1 System Architecture Overview

Med Sum-AI is structured as a four-stage end-to-end pipeline. Stage 1 (Data Preprocessing) covers section parsing, de-identification verification, tokenization, and train/validation/test splitting (70/15/15). Stage 2 (NLP Summarization) fine-tunes and evaluates BERT, BioBERT, ClinicalBERT, and BART, implements an ensemble voting layer, and applies hallucination detection and mitigation. Stage 3 (Feature Extraction and Prediction) parses 20 structured features from NLP summaries and CASIA-CXR label files, trains Logistic Regression, Random Forest, XGBoost, and Cox PHM, and generates pathology classification predictions for the five CASIA-CXR disease categories. Stage 4 (Explainability and Reporting) generates SHAP token attribution

maps, LIME feature importance charts, administers the clinician trust survey, and presents results via a Gradio web dashboard.

5.2 RQ1 — NLP Model Benchmarking

All four transformer models are fine-tuned on the same CASIA-CXR dataset split to ensure comparability. The Impressions section of each radiology report serves as the gold-standard reference summary for evaluating abstractive outputs. BERT-based extractive summarization (BERTSum) uses binary sentence classification (include/exclude) with an Adam optimizer (lr = 2e-5, batch = 16, 5 epochs). BioBERT and ClinicalBERT are fine-tuned using the same extractive framework; ClinicalBERT benefits from clinical domain pre-training (Alsentzer et al., 2019) and is further fine-tuned on CASIA-CXR radiology reports. BART abstractive fine-tuning uses facebook/bart-large-cnn as the base model, cross-entropy loss on source-summary pairs, and beam search decoding (k = 4, length penalty = 2.0). A weighted voting ensemble selects the final output per document based on validation ROUGE-L scores.

Statistical analysis: One-way ANOVA compares ROUGE-L scores across the four model groups at alpha = 0.05. Tukey HSD post-hoc test identifies specific pairwise differences. Effect size is quantified using Eta-squared (η^2). Bootstrapped 95% confidence intervals (B = 1,000 resamples) are reported for all ROUGE and BERTScore estimates. Expected finding: ClinicalBERT fine-tuned on CASIA-CXR radiology reports will achieve ROUGE-L $\geq$ 0.46, outperforming general BERT (expected ROUGE-L $\leq$ 0.38) due to its clinical domain vocabulary alignment.

5.3 RQ2 — Clinical Expert Validation

Clinical quality is assessed using a five-dimension, five-point Likert rubric covering factual accuracy, completeness, conciseness, clinical relevance, and readability (maximum score = 25 per summary). Three board-certified clinicians independently score 60 summaries per model (240 total across all four models). Inter-rater reliability is measured using the Intraclass Correlation Coefficient (ICC; two-way mixed model), with ICC > 0.75 as the acceptability threshold for good reliability. Group comparison across models uses the Kruskal-Wallis H-test (appropriate for ordinal rubric scores) with Dunn's post-hoc pairwise test. Predicted outcome: BART abstractive summaries will score $\geq$ 3.8/5.0 on factual accuracy and $\geq$ 4.0/5.0 on conciseness.

5.4 RQ3 — Hallucination Detection and Mitigation

Hallucination is operationally defined as any generated statement not supported by the source clinical note. Automated faithfulness scoring uses Fact CC (a fine-tuned BERT-based factual consistency classifier) and QuestEval (a QA-based metric). Manual annotation is performed by two independent clinical annotators on 150 summaries, with Cohen's Kappa quantifying inter-annotator agreement (target $\kappa > 0.70$). Hallucination rate is calculated as: hallucinated statements / total generated statements per model.

Three mitigation strategies are compared against the baseline BART beam search output: (A) constrained decoding, where clinical entity names (diseases, drugs) are constrained to appear only if present in the source; (B) a copy mechanism integrated into the BART decoder (pointer-generator style); and (C) RAG (Retrieval-Augmented Generation), where FAISS vector search retrieves the three most similar clinical notes to condition BART generation on retrieved context. McNemar's test is used to compare hallucination rates before and after applying each mitigation

strategy. Prediction: RAG-enhanced BART will reduce hallucination rate by ≥ 40% compared to baseline BART.

5.5 RQ4 — Predictive Outcome Modelling

Feature engineering produces 20 structured variables from the CASIA-CXR radiology report text and label files, spanning four categories: NLP-derived from BART summaries (sentiment polarity, clinical entity count, finding severity score, diagnostic term frequency, impression length), report structural features (word count, sentence count, Findings-to-Impressions ratio, presence of measurement terms, negation count), image metadata (pathology category label, multi-label co-occurrence flag), and linguistic complexity (Flesch-Kincaid readability, medical abbreviation density, numerical value count, uncertainty hedge count, passive voice ratio, temporal marker count, recommendation presence flag).

Four statistical predictive models are deployed. Model 1 is Logistic Regression (L2-regularized, C = 1.0) for binary pathology classification per CASIA-CXR category, with outputs including coefficient estimates, odds ratios, and 95% CIs via bootstrap; Box-Tidwell test for log-odds linearity and VIF < 10 for multicollinearity. Model 2 is Random Forest (n_estimators = 500, max_depth = None, min_samples_leaf = 5) for multi-category pathology classification, with feature importance via permutation importance and out-of-bag error as secondary validation. Model 3 is XGBoost (max_depth = 6, learning_rate = 0.05, n_estimators = 300) for all five pathology categories (one-vs-rest), with hyperparameters tuned via Bayesian optimization and SHAP values for global and local feature attribution. Model 4 is the Cox Proportional Hazard Model applied to radiology severity score as a time-to-event analog, with Schoenfeld residual validation of the proportional hazards assumption and Kaplan-Meier survival curves stratified by predicted risk quartile. A Linear Regression baseline (OLS with HC3 robust standard errors) is fitted for continuous severity score prediction, with Shapiro-Wilk, Breusch-Pagan, Durbin-Watson, and VIF assumption checks.

Model performance is evaluated using the metrics presented in Table 3 and Table 4 below. Calibration is assessed using the Hosmer-Lemeshow goodness-of-fit test and reliability diagrams; Platt scaling or isotonic regression is applied post-hoc if models are poorly calibrated.

Table 3. NLP Summarization Evaluation Metrics with Formulae

Metric	Formula	Target	Rationale
ROUGE-1	R1 = |Unigrams(cand) ∩ Unigrams(ref)| / |Unigrams(ref)|	> 0.45	Unigram overlap with reference
ROUGE-2	R2 = |Bigrams(cand) ∩ Bigrams(ref)| / |Bigrams(ref)|	> 0.22	Phrasal overlap quality
ROUGE-L	RL = LCS(candidate, reference) / Len(reference)	> 0.40	Longest common subsequence
BERTScore-F1	F1 = 2·(BERTPrec·BERTRec) / (BERTPrec + BERTRec)	> 0.88	Semantic similarity via contextual embeddings
Fact CC	P(consistent | claim, document) via fine-tuned BERT classifier	> 0.80	Factual consistency (anti-hallucination)

QuestEval	F1 over QA pairs generated from source and candidate	> 0.70	Information overlap via QA approach

Table 4. Predictive Model Evaluation Metrics with Formulae

Metric	Formula	Target	Use Case
AUC-ROC	Area under ROC curve: $\int$ TPR d(FPR)	> 0.80	Discrimination — all classifiers
Brier Score	$BS = (1/N) \Sigma (f_t - o_t)^2$	< 0.15	Calibration quality
C-statistic	Concordance index for Cox PHM: $P(T_i < T_j \mid Z_i > Z_j)$	> 0.75	Cox model discrimination
RMSE (Severity)	$RMSE = \sqrt{[(1/N) \Sigma (\hat{y}_i - y_i)^2]}$	< 3.0	Severity score regression accuracy
Matthews CC	$MCC = (TP \cdot TN - FP \cdot FN) / \sqrt{[(TP+FP)(TP+FN)(TN+FP)(TN+FN)]}$	> 0.60	Balanced binary classification
H-L Test	$\chi^2 = \Sigma (O_g - E_g)^2 / [E_g(1-E_g/n_g)]$ over 10 groups	p > 0.05	Calibration goodness-of-fit

5.6 Statistical Hypotheses Summary (RQ1–RQ4)

The following hypotheses are formally testable predictions grounded in published clinical NLP and predictive modelling benchmarks. Each will be evaluated at alpha = 0.05 using the designated statistical test.

Table 5. Formal Statistical Hypotheses for RQ1–RQ4

RQ	Hypothesis	Predicted Result	Statistical Test
RQ1	H1: Clinical BERT > BERT	Clinical BERT ROUGE-L ≥ 0.46; BERT ROUGE-L ≤ 0.38	One-way ANOVA + Tukey HSD post-hoc
RQ1	H2: BART ≥ extractive models	BART BERTScore-F1 ≥ 0.89; outperforms extractive on coherence	Paired t-test on BERTScore
RQ2	H3: Abstractive rubric ≥ 3.8/5	BART clinical accuracy ≥ 3.8; readability ≥ 4.0	Kruskal-Wallis + Dunn post-hoc
RQ3	H4: RAG reduces hallucination	RAG-BART hallucination rate ≤ 5% vs. ≥ 12% baseline	McNemar's test (before/after)
RQ4	H5: XGBoost AUC ≥ 0.82 (pathology classification)	NLP-feature XGBoost outperforms LR by ≥ 0.08 AUC for pathology classification	DeLong test for AUC comparison
RQ4	H6: Cox PHM C-stat ≥ 0.75	Cox PHM identifies top severity-risk quartile with HR ≥ 2.5	Log-rank test on K-M curves

	(severity progression)		
RQ5	H7: SHAP increases trust	Mean TAM trust score increases ≥ 1.2 points (7-point scale)	Wilcoxon Signed-Rank Test
RQ6	H8: GI(Clinical BERT) > 0.85	Clinical BERT generalizes to radiology (GI ≥ 0.85); general BERT GI < 0.70	Paired t-test CASIA-CXR vs. i2b2

6. Recommendations and Applications

6.1 Target End Users and Benefits

Med Sum-AI is designed for use by a diverse range of healthcare stakeholders. Attending physicians and residents are primary users, benefiting from instant 150-word summaries and risk-stratified pathology probability scores during morning rounds and patient handovers. Radiologists and emergency teams benefit from prioritized, section-structured radiology report summaries with real-time pathology probability scores to support rapid diagnostic decisions. Hospital administrators can use standardized structured summaries for compliance, billing, and utilization management, while pathology classification predictions enable proactive, targeted diagnostic interventions to reduce missed diagnoses and associated penalties.

Hospital information systems (HIS) and EHR integration teams are served by the open-source, Docker containerized architecture, which is designed for modular FHIR-compliant API integration with Epic and Cerner platforms. Clinical researchers gain automated cohort extraction and case summarization capabilities across the five CASIA-CXR pathology categories, eliminating laborious manual radiology report review. Healthcare AI developers gain a fine-tuned, open-source Med Sum-AI codebase as a starting point for institutional adaptation and further research.

6.2 Deployment Vision and Social Impact

The deliverable is a three-component system: (1) a Python/Hugging Face pipeline for radiology report summarization; (2) a scikit-learn/XGBoost predictive module for outcome scoring; and (3) a Gradio web interface presenting, side-by-side, the source clinical report, the generated summary, the SHAP token heatmap, and the predicted risk score with LIME explanation. The system is containerized in Docker for reproducible deployment across institutional environments. All components are released under the MIT license on GitHub.

If deployed at scale, Med Sum-AI could save each physician 1–2 hours of documentation and review time per shift — equivalent to over 250,000 clinician-hours annually across a 500-bed hospital system. More importantly, accurate pathology classification from NLP-extracted radiology report features could enable faster, consistent diagnostic decision support, with the potential to reduce diagnostic error rates by an estimated 10–15%, generating substantial improvements in both patient outcomes and institutional financial performance.

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Appendix A — Statistical Techniques Summary

Table A1 provides a complete index of all statistical and machine learning techniques applied across the six research questions, including technique type, the research question each addresses, and its analytical purpose. This appendix supplements Section 5 by providing a consolidated reference for the evaluator.

Table A1. Complete Statistical and ML Techniques by Research Question

Technique	Type	Applied to RQ	Purpose
One-Way ANOVA	Inferential — Parametric	RQ1	Compare ROUGE-L across four NLP models
Tukey HSD Post-Hoc	Multiple Comparison	RQ1	Pairwise model performance differences
Bootstrapped 95% CI	Resampling	RQ1, RQ4	Robust uncertainty quantification (B = 1,000)
Kruskal-Wallis H-Test	Inferential — Non-Parametric	RQ2	Compare ordinal rubric scores across models
Intraclass Correlation (ICC)	Reliability Analysis	RQ2	Inter-rater agreement among clinical experts
Cohen's Kappa	Agreement Statistic	RQ3	Hallucination annotation agreement
McNemar's Test	Inferential — Paired Proportions	RQ3	Before/after hallucination rate comparison
Logistic Regression (L2)	Statistical ML — Classification	RQ4	Pathology classification prediction (baseline)
Random Forest	Ensemble ML — Classification	RQ4	Non-linear feature interaction modelling
XGBoost	Gradient Boosting	RQ4	Best-performance outcome prediction
Cox Proportional Hazard Model	Survival Analysis	RQ4	Time-to-event modelling (severity score progression)
Linear Regression (OLS)	Statistical — Continuous Outcome	RQ4	Severity score continuous prediction baseline
Hosmer-Lemeshow Test	Goodness-of-Fit	RQ4	Calibration testing for binary classifiers
DeLong Test	AUC Comparison	RQ4	Statistical comparison of AUC-ROC curves
Wilcoxon Signed-Rank	Non-Parametric Paired	RQ5	Trust score change before/after SHAP exposure
Kaplan-Meier Curves	Descriptive Survival Analysis	RQ4	Visualize severity risk progression by group
Schoenfeld Residuals	Assumption Testing	RQ4	Validate proportional hazards assumption (Cox)

VIF / Breusch-Pagan	Assumption Testing	RQ4	Multicollinearity and homoscedasticity checks
Paired T-Test	Inferential — Parametric Paired	RQ6	Domain generalization ROUGE-L comparison